

## Fourteen Caption Users

AP Statistics · Unit 3 · Topic 3.2 · B: 2026-12-01 · E: 2026-12-02 · TEACHER KEY

### CED TETHER

- **Skill 3.B** — Determine parameters for probability distributions.
- **Skill 3.C** — Describe probability distributions.
- **Skill 4.B** — Interpret statistical calculations and findings to assign meaning or assess a claim.
- **EU UNC-3** — Probabilistic reasoning allows us to anticipate patterns in data.
- **LO UNC-3.K** — Determine parameters of a sampling distribution for sample proportions. [Skill 3.B]
- **EK UNC-3.K.1** — For independent samples of a categorical variable from a population with proportion  $p$ , the sampling distribution of the sample proportion has mean  $\mu = p$  and a standard deviation  $\sigma = \sqrt{p(1-p)/n}$ .
- **EK UNC-3.K.2** — If sampling without replacement, the standard deviation of the sample proportion is smaller than the formula above. If the sample size is less than 10% of the population size, the difference is negligible.
- **LO UNC-3.L** — Determine whether a sampling distribution for a sample proportion can be described as approximately normal. [Skill 3.C]
- **EK UNC-3.L.1** — For a categorical variable, the sampling distribution of the sample proportion will have an approximate normal distribution provided the sample size is large enough:  $np \geq 10$  and  $n(1-p) \geq 10$ .
- **LO UNC-3.M** — Interpret probabilities and parameters for a sampling distribution for a sample proportion. [Skill 4.B]
- **EK UNC-3.M.1** — Probabilities and parameters for a sampling distribution for a sample proportion should be interpreted using appropriate units and within the context of a specific population.

**Essential question:** How do sampling, spread, and shape conditions limit what a surprising sample proportion can show?

**DOK-3 skill:** 4.B · **FRQ pattern:** failed-large-counts-normal-tail-vs-valid-binomial-and-bias

**DOK rationale:** Part (c) separates independence, shape, and unbiasedness to evaluate two overclaims and use a supplied probability to bound the conclusion supported by a random sample.

**First take** → rules + (a)–(c) → turn in

**DOK:** 1 → 3

**Minutes:** 45

#### Teacher does

- Board slide up at the bell. Ask students to mark which sentence in each claim is about shape, center, or evidence.
- At minute 5, read the rules box aloud once; students start (a).
- Return to the board slide at minute 33. Circulate and press the distinct roles of 10 percent and Large Counts.
- Collect at the door. Score part (c) only, E/P/I.

#### Students do

- Choose whose reasoning is closer and name one condition before the lesson.
- Use the rules box to find the sampling mean and standard deviation and check both conditions.
- Finish (a)–(c), revise the first take in the exit line, and turn in the sheet.

#### Questions to ask

- Which condition supports treating the 100 selections as approximately independent?
- Why does  $np = 8$  address shape rather than the estimator's long-run center?
- What values of  $X$  belong to  $\hat{p} \geq 0.14$ ?
- How do 0.0135 and about 0.028 change the strength, but not the direction, of the evidence?

#### Adult role

Solo teacher. Circulate 5 pairs. Require students to attach each condition to its job and to distinguish evidence against 0.08 from proof of a changed population rate.

#### [WARN] LOOK FOR

- Applying an  $n > 30$  rule carried over from sample means.
- Calling a nonnormal sampling distribution biased.
- Calculating exactly 14 rather than 14 or more caption users.

#### SCORING PART (c) — E / P / I

#### Expected elements

- **judgment-1** (required) — Uses the passing 10 percent condition and failed  $np=8$  normal-shape condition to distinguish the two model requirements.
- **judgment-2** (required) — Uses supplied tail near 0.0282 as unusual evidence under  $p=0.08$ , without claiming proof of an increase.
- **judgment-3** (required) — Explains that SRS unbiasedness concerns center, whereas normality concerns shape, and rejects both overclaims.

|          |                                                                                                    |
|----------|----------------------------------------------------------------------------------------------------|
| <b>E</b> | Meets all three checks with correct contextual reasoning; equivalent wording is acceptable.        |
| <b>P</b> | Shows at least one substantive correct check, but has an omission or error that prevents E.        |
| <b>I</b> | Shows none of the three checks with substantive correct reasoning; a choice alone is insufficient. |

#### Common mistakes

- Checking only  $n > 30$  instead of both expected counts
- Using  $P(X = 14)$  rather than the upper-tail event  $P(X \geq 14)$
- Saying the 10 percent condition determines normal shape rather than approximate independence and spread
- Equating a skewed sampling distribution with a biased estimator

**[STAR] TODAY'S PROBLEM — annotated**

Suppose a streaming platform has 5,000 active accounts and says that  $p = 0.08$  of them regularly use captions. An SRS of 100 accounts, selected without replacement, contains 14 regular caption users, so  $\hat{p} = 0.14$ .

**Jules:** “Because  $n = 100 > 30$ ,  $\hat{p}$  is approximately normal. I get  $P(\hat{p} \geq 0.14) = 0.0135$ , which proves caption use increased.”

**Nora:** “Because  $np = 8 < 10$ , a normal model is not justified. Therefore  $\hat{p}$  is biased and this sample tells us nothing.”

A binomial calculator gives  $P(X \geq 14) = 0.0282$  for  $X \sim \text{Bin}(100, 0.08)$ . This closely approximates sampling from 5,000 accounts.

**Caption-use sample**

| Source     | Accounts | Users | Rate |
|------------|----------|-------|------|
| Population | 5,000    | 400   | 0.08 |
| SRS        | 100      | 14    | 0.14 |

**FIRST TAKE (5 min)**

Does finding 14 caption users instead of the expected 8 prove that use increased? Give one reason.

**Key:** Not graded. Each student has one correct observation and one overreach.

*Rules callout “CHECK INDEPENDENCE, THEN SHAPE (keep on the page)” is printed on the student sheet.*

**DOK 1** (a) Identify  $p$  and verify the sample proportion  $\hat{p} = 14/100$ .

**Key:** The parameter is  $p$ , the proportion of all 5,000 active accounts that regularly use captions. The statistic is the sample proportion  $\hat{p} = 14/100 = 0.14$ . Thus  $N = 5,000$  and  $n = 100$ .

**DOK 2** (b) Find the mean and standard deviation of the sampling distribution of  $\hat{p}$ . Check the 10 percent condition and both Large Counts values.

**Key:** The sampling distribution has mean  $\mu_{\hat{p}} = p = 0.08$  and standard deviation  $\sigma_{\hat{p}} = \sqrt{0.08(0.92)/100} = 0.027129 \dots \approx 0.0271$ . The 10 percent condition holds because  $100 < 0.10(5,000) = 500$ , so dependence from sampling without replacement is negligible. Large Counts does not fully hold:  $np = 100(0.08) = 8 < 10$ , while  $n(1 - p) = 100(0.92) = 92 \geq 10$ .

**DOK 3** (c) In 3–4 sentences, explain what each student gets right and wrong. Use the condition checks and the supplied probability 0.0282. State what the sample suggests about caption use and why a failed normal-shape check does not make the estimate biased.

**Key:** Neither claim is fully correct: the sample is random and  $100 < 500$ , but  $np = 8 < 10$ , so Nora correctly rejects the normal approximation. The supplied binomial tail, about 0.0282, makes 14 or more unusual under  $p = 0.08$ , giving evidence worth investigating rather than proof of an increase. The SRS produces an unbiased sample proportion; the failed condition concerns shape, not its long-run center. Jules overstates the evidence and Nora wrongly discards a useful estimate.

**EXIT**

One thing the rules box changed about my first take: \_\_\_\_\_